Weeks 5–6 | **CLO** 4 (+1) | **Midterm** 3/11/2026 · 6:15–7:15 pm | **Target** A ≥ 90 | **File** 1 / 11

00 What the outline actually tests

ما الذي يطلبه الكورس أوت لاين فعليًا من هذا الفصل

Chapter 4 is not graded on how much you can recite. The outline pins it to one learning outcome with three assessment criteria, and every quiz, midterm and final question is built from those criteria. Read this box first; it tells you where the marks live.

CHAPTER 4 IN THE COURSE SCHEDULE

WEEK	SECTIONS DELIVERED	ASSESSMENT ATTACHED
5	4-1 Sample Spaces and Probability 4-2 The Addition Rules 4-3 Multiplication Rules & Conditional Probability *Lab 5 — RStat2101: Descriptive Measures*	Exercises Ch. 4, p. 247 `{6, 8, 16, 40}` Moodle Exercises **Lab Quiz I**
6	4-4 Counting Rules 4-5 Probability and Counting Rules	Moodle Exercises (cont.) **In-Class Quiz II**
8	—	**Midterm Exam** · 25% · 3/11/2026, Tue 6:15–7:15 pm · **Chapters 1, 2, 3, 4**
16	—	**Final Exam** · 45% · 29/12/2026, 8:00–11:00 am · all chapters

CLO4 — Apply Fundamental Probability Concepts

This is the outcome Chapter 4 serves. Its assessment criteria, verbatim from the outline:

#	CRITERION — "TO ACHIEVE THIS OBJECTIVE, THE STUDENT MUST…"	COVERED BY
A	**Calculate probabilities using counting rules and basic probability principles.**	**All of Chapter 4.** This is the one that matters here.

#	CRITERION — "TO ACHIEVE THIS OBJECTIVE, THE STUDENT MUST…"	COVERED BY
B	Distinguish between discrete and continuous probability distributions.	Chapter 5 — not Chapter 4.
C	Compute and interpret expected value and variance for discrete random variables.	Chapter 5 — not Chapter 4.

CLO4 is assessed by **Tutorial Questions + In-Class Quiz + Active Learning Exercise + Midterm Exam + Final Exam**. That is five separate hits. CLO1 (*Identify key statistical concepts and terms*) also runs through this chapter: definition questions — *what is a sample space, what does mutually exclusive mean* — are CLO1 marks and are the cheapest marks in the course.

THE SINGLE HIGHEST-VALUE SENTENCE IN THIS DOCUMENT

Criterion 4-A says "**counting rules and basic probability principles**" in the same breath. That is the examiner telling you in advance that §4-5 problems — where you compute `n(E)` and `n(S)` with combinations and *then* divide — are the signature Chapter 4 question. Sections 4-4 and 4-5 carry In-Class Quiz II on their own. Do not treat counting as an afterthought at the end of the chapter; it is the destination of the whole chapter.

خلاصة سريعة

الفصل الرابع كله يخدم مخرج تعلم واحد (CLO4-A): حساب الاحتمالات باستخدام قواعد العدّ والمبادئ الأساسية. يُقاس في خمسة تقييمات: التمارين، الكويز الصفي، نشاط التعلّم النشط، الميدتيرم (25%)، والنهائي (45%). الأقسام **4-4** و **4-5** (قواعد العدّ) تحمل **الكويز الصفي الثاني** بالكامل — لا تؤجلها.

Grade arithmetic: what A actually costs

The A–F scale puts **A at ≥ 90** and A− at ≥ 86. With In-Class Quizzes 15%, Lab Quiz 10%, Active Learning 5%, Midterm 25% and Final 45%, Chapter 4 content appears in at least four of those five buckets. A clean Chapter 4 is worth more than any single other chapter in the first half of the course, because it is the last chapter on the midterm *and* the foundation of Chapters 5 and 6 on the final.

لماذا هذا الفصل مهم لدرجتك؟

الـ A تبدأ من 90 . الفصل الرابع يدخل في أربعة من خمسة تقييمات على الأقل، وهو **آخر فصل في الميدتيرم**، و**الأساس الذي يُبنى عليه الفصلان الخامس والسادس** في النهائي (التوزيع ذو الحدين والتوزيع الطبيعي كلاهما احتمالات). أي ضعف هنا ينتقل معك إلى نهاية الفصل الدراسي، وأي إتقان هنا يسهّل عليك ما بعده.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests ← this file

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.